Project Proposal version 2
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Lane: development

Framing Formula

For children aged 0–59 months in Sindh, predict whether the child is stunted (height-for-age z-score < −2), using PDHS 2017-18 microdata, only variables an LHW can collect by asking (child age and sex, mother's education, asset-based household wealth, birth order and interval, household size, water/sanitation, rural/urban, antenatal visits, feeding practices), trained on all provinces **(using the standard survey weights, which exclude AJK and GB)** and tested on held-out Sindh clusters, **with the target** recall ≥ 0.75 at precision ≥ 0.60, so that the senior manager of the Sindh Lady Health Workers Programme can take the decision to send LHWs to flagged households first for length measurement and nutrition counselling, instead of visiting in routine order.

What changed from Week 1

After working with the data, four things changed or were tested. First, the usable sample is much smaller than I assumed: only 806 of 2,278 Sindh children have a height-for-age score, and 85 children nationally carry the flagged code 9998 (Sindh share not yet known), which I treat as missing.

Second, I had assumed missing children were mostly absent or unmeasured on interview day. The measurement-result column showed otherwise: 7,490 of 8,482 missing scores are labelled “no measurement found in household”, and only 111 children were absent and 86 refused.

Third, rows are not independent, because the same caseid appears with different midx, so siblings share a mother. This supports my plan to hold out whole clusters rather than random rows.

Fourth, with about 800 measured Sindh children, the held-out test set will be small, so recall and precision will be noisy. The success metric and the baseline rule are unchanged because I have not tested them on the data yet.

These findings led to two edits in the proposal. In the framing formula, the model is now trained on all provinces using the standard survey weights, which exclude AJK and GB, and recall ≥ 0.75 at precision ≥ 0.60 is stated as a target. In the sampling frame, the Sindh sample size that v1 promised to count in Week 2 is now reported.

Stakeholder

Who commissioned this work? Whose problem is it?

The head of the Sindh Lady Health Workers Programme, Government of Sindh. The programme runs over 20,898 LHWs and 723 Lady Health Supervisors across Sindh, yet Sindh's stunting rate was 50% in the PDHS 2017-18 factsheet, and LHW visit capacity is limited. Their problem is deciding which households LHWs should reach first for length measurement and nutrition counselling, and they would use an LHW risk-screening app to do it.

What decision are they making today (without your model)?

My assumption: LHWs visit households in routine rounds (monthly home visits), and planning relies on province/district-level stunting rates, with no household-level ranking.

What organization / team do they sit in?

Lady Health Workers Programme, Government of Sindh (Health Department).

Users

Who actually interacts with the output of your model?

Lady Health Workers (each oversees about 150 to 200 households), with Lady Health Supervisors reviewing.

How will the user see the output?

A simple app: the LHW enters the answers she already collects for a child, and gets a risk score with a red/amber/green flag. A ranked list view sorts a catchment's households by risk. The LHW can override a flag but must write a one-line reason.

Who is affected downstream by the decision?

Children under 5 and their mothers in flagged households (earlier measurement and counselling). A false flag costs a family one extra visit, and a missed stunted child gets no early priority.

Problem Type

Classification (binary: stunted or not). Sindh's stunting rate was 50% in the PDHS 2017-18 factsheet, so accuracy alone is meaningless (flagging everyone already scores 50%).

Sampling Frame

PDHS 2017-18, a probability sample (two-stage cluster sampling, so survey weights matter). It gives reliable estimates for 4 provinces, which includes Sindh.

Who does it miss?

- Children not measured or absent on interview day. **Only 806 of 2,278 Sindh children have a height-for-age score.** Published analyses of measured children used 4,226 and 4,499 children nationally.
- Post-2018 shocks (e.g. the 2022 floods hit Sindh badly), so the model needs re-validation on fresh data before real use.
- LHW outreach is mainly rural populations and urban slum communities, while the survey covers all households, so I will report performance separately by rural/urban and wealth group.
- It is a snapshot, so I predict current stunting, not future stunting.

Baseline

What would a stupid rule-of-thumb predictor get?

- Flag everyone: recall 1.00, precision ≈ 0.50 in Sindh.
- One-line rule: flag = (wealth_quintile <= 2) & (mother_education == "none"). I will compute its actual recall and precision in Week 2.

Cost of Error

Cost of a false positive: A non-stunted child is flagged, so the LHW makes one unneeded visit (assumed ~30–45 min). Small and recoverable.

Cost of a false negative: A stunted child is missed and loses the early-intervention window (stunting is largely irreversible after ~age 2). Assumption for v1: FN ≈ 10× FP.

Therefore, my metric optimizes for: recall (≥ 0.75), subject to a precision floor of 0.60. The floor is there because flagging everyone already gives recall 1.00 at precision 0.50, but LHWs cannot visit everyone first, so a list only helps if it is meaningfully better than random.